

# An-Sheng (Anson) LU

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## EDUCATION

**University of California, Riverside, CA, USA** | *Expected Spring 2029* **2025 – Present**

Major in Electrical Engineering at Department of Electrical and Computer Engineering | Cumulative GPA: 3.80.

**Relevant Coursework:** Multivariable Calculus, Physics, Data Structures & Algorithms (C++).

**Scholarship:** UCR Achievement Scholarship (2025–29).

**Victoria Academy, Yunlin, Taiwan**

**2022 – 2025**

IB Bilingual Diploma, July 6, 2025 | GPA: 3.94 (IBDP) / 3.86 (Cumulative).

– International Baccalaureate Diploma Programme (IBDP) HL subjects: Mathematics AA (7), Physics (6), Chemistry (6).

**Graduation Awards:** Chairman's/Mayor's Award, Dr. Cho Rui-Chiao Scholarship, Excellence Award in Mathematics.

## EXPERIENCES

**Intern, Electrical Subsystems, Highlander Racing (FSAE), University of California, Riverside** **Apr. 2026 – Present**

- Contributed to the design of an industrial automation PCB in Altium Designer, from schematic capture of analog sensing stages to routing power stages.

**Research Assistant, Robotics and Medical Systems (RaMS) Lab, University of California, Riverside** **Oct. 2025 – Present**

- Independently developed a CV/DL pipeline using AISFormer in collaboration with the UCR Entomology Dept., automating biomass estimation and growth stage classification for mealworms; presented findings in biweekly meetings with PI.
- Architected automated workflows featuring normal distribution filtering and sensor calibration to translate image data into high-accuracy phenotypic measurements for life-cycle analysis.

**Inventor, Taiwan Intellectual Property Office, Taiwan**

**June 2024 – Aug. 2025**

- Patent: *Baton Device with Directional Indication* (Invention Patent No. I894025). [Taiwan Patent Search System](#).
- Independently designed a signaling device with custom hardware logic using an ESP32 and MPU-6050 (IMU) to process real-time indication; managed conceptual design and circuit prototyping phases through to formal patent approval.

**Intern, Robotics and Mechatronics Center, German Aerospace Center (DLR), Oberpfaffenhofen, Germany** **Aug. 1 – 30, 2024**

- Integrated a custom robotic system using an OpenManipulator-X arm; collaborated with advisors on system design and implemented 3D-printed structural interfaces and motion control.
- Modeled mass-spring-damper systems in MATLAB/Simulink to analyze resonance and power dissipation; designed a critically damped feedback control system to optimize stability.
- Designed and assembled a 3x3x3 LED matrix using an Arduino Micro; implemented a 4-mode firmware system featuring real-time pace control via ADC and a variable resistor.

**Robotics Club Member, National Yunlin Univ. of Sci. & Tech. (NYUST), Yunlin, Taiwan**

**Sep. 2023 – June 2024**

- Collaborated with club members to design an omnidirectional mobile robot with a three-Mecanum wheel chassis; executed sensor fusion using Ultrasonic and Color Detection sensors for obstacle avoidance and environmental interaction via LabVIEW.
- Implemented basic ROS interfacing for system communication.

## HONORS & AWARDS

- 1st Place Winner, Robotics Competition, Yunlin, Taiwan, Dec. 25, 2023.
- 1st Place Winner, Industrial Robot Competition, Kaohsiung, Taiwan, Mar. 9, 2024.
- Final Shortlisted Prize, Electrical Engineering Category, 2024 All Ring Competition, Tainan, Taiwan, Oct. 24, 2024
- Victoria Academy Scholarships (5x recipient, 2022–2024).

## INDEPENDENT ACADEMIC STUDIES

- Determination of gravity vector** (Oct. 2024): Identified gravity vector in body-fixed frame using tri-axial accelerometer.
- Resonance in a shaking device** (Mar. 2025): Analyzed power dissipation changes with spring constant variations.

## SKILLS

- Language:** Mandarin (Native), English (Fluent), Taiwanese/Min Nan Chinese (Conv.), German (Basic)
- Programming:** C/C++, Python (PyTorch, OpenCV), LabVIEW, MATLAB/Simulink, ROS (Basic).
- Software & Hardware Skills:** Tinkercad, Creo, Arduino (IDE), Visual Studio/VS Code, Altium Designer, 3D Printing.
- Interest:** Basketball (Clippers), Formula 1 (Red Bull Racing), Strategic Games (Chess/Xiangqi), Robotics, Electronics.